BTS-I&II(R/S)-(04/09)-21-0805 Reg. No. | | | | | | | | |

B

B.Tech. Degree I&II Semester Regular/SupplementaryExamination April/ September 2021

CE/EE 19-200-0103A ENGINEERING GRAPHICS

(2019 Scheme)

(Regular)

SCHOOL OF ENGINEERING, KOCHI-682 022 COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY * LIBRARY *

Time: 3 Hours Maximum Marks: 60

PART A

(1 × 12 = 20)

I. On a building plan, a line 20 cm long represents a distance of 10 m. Devise a diagonal scale for the plan to read upto 12 m, showing meters, decimetres and centimetres. Show on your scale the lengths of 6.48 m and 11.14 m. (12)

OR

II. Draw an Involute of a circle of 40 mm diameter. Also draw a normal and tangent to the point 100 mm from the centre of circle. (12)

PART B

(4 × 15 = 60)

III. (a) A straight line AB 65 mm long has its end A 15 mm in front of VP and 40 mm above HP while the other end B is 30 mm in front of VP and 20 mm above HP. Draw its projections. (6)

(b) The end projectors of line AB is 60 mm apart. A is 20 mm above HP and 30 mm in front of VP and end B is 70 mm above HP and 45 mm behind VP. Find the true length and true inclinations of the line. (9)

OR

IV. (a) Draw the projections of a rhombus having diagonals 125 mm and 50 mm long, the smaller of which is parallel to both the principal planes, while the other is inclined at 30^0 to the HP. (6)

(b) Draw the projections of a regular hexagon of 25 mm side, having one of its sides in the HP and inclined at 60^0 to VP and its surface making an angle of 45^0 with the HP. (9)

V. A pentagonal pyramid edge of base 25 mm and height 60 mm rests on a corner of its base in such a way that the slant edge containing the corner makes an angle of 45^0 with HP and 30^0 with VP. Draw its projections. (15)

OR

VI. A pentagonal pyramid, having base with a 30 mm side and 70 mm length, is resting on its base in the HP with an edge of the base parallel to the VP. It is cut by a section plane perpendicular to the VP, inclined at $60^0$to the HP and bisecting the axis. Draw its front view and sectional top view and true shape of the section. (15)

VII. Hexagonal pyramid 30 mm base edge and 60 mm height is resting on its base on HP with one of its base edges parallel to VP. It is cut by a sectional plane, inclined 30^0 to HP and passing through a point on axis 25mm above its base. Draw the development. (15)

OR

VIII. A square prism with a 40 mm base side and 100 mm long axis is resting on its base on HP with a side of base parallel to VP. It is penetrated by a horizontal cylinder with a 50 mm base diameter and 100 mm long axis such that their axes bisect each other at right angles. Draw the three views of the combination and show the curves of intersection. (15)

(P.T.O)

IX. A rectangular slab 75 mm × 50 mm × 20 mm is surmounted by a cube of 40 mm size. On the top of the cube, rests a square pyramid of altitude 40 mm and side of base 25 mm. The axes of the solids are in the same straight line. Draw the isometric view of the solid. (15)

OR

X. A rectangular prism 25 mm × 30 mm side and 50 mm long is lying on the ground plane on one of its rectangular faces in such a way that one of its end face is parallel to and 10 mm behind the picture plane. The station point is 60 mm away from the axis of the prism towards the left. Draw the perspective view of the prism if the station point is located 55 mm infront of the picture plane and 40 mm above the ground plane. The prism is resting on the ground plane on its 50 mm × 25 mm rectangular face. (15)

***